

VINEET PUNYAMOORTY

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EDUCATION

Purdue University <i>PhD in Electrical and Computer Engineering</i> <i>Research Areas: Machine Learning, Multimodal Learning, Computer Vision, Generative Models</i>	West Lafayette, IN Aug 2022 – present
Indian Institute of Technology (IIT) Bombay <i>Bachelor of Technology & Master of Technology</i> Major: Electrical Engineering	Mumbai, India Aug 2015 – May 2020

SELECTED PUBLICATIONS

- [1] Dynamic Obstacle Avoidance through Uncertainty-Based Adaptive Planning with Diffusion
V. Punyamoorthy, et al. **IROS 2025**. [[ArXiv](#)]
- [2] Contrastive Cross-Modal Learning for Infusing Chest X-ray Knowledge into ECGs
V. Punyamoorthy, et al. *Under Review* at **IEEE Journal of Biomedical and Health Informatics**. [[ArXiv](#)]
- [3] CarE-X: ECG-based Prediction of Cardiomegaly Metrics through Alignment with X-rays
D. Tamboli, **V. Punyamoorthy**, A. Malusare, and V. Aggarwal. *Under Review* at **IEEE Journal of Biomedical and Health Informatics**.
- [4] Augmenting Generative Models with Biomedical Knowledge Graphs Improves Targeted Drug Discovery
A. Malusare, **V. Punyamoorthy**, V. Aggarwal. *Under Review* at **IEEE Transactions on Artificial Intelligence**.
- [5] Privacy-aware Image Editing and Enhancement Pipeline
D. Tamboli, **V. Punyamoorthy**, V. Aggarwal. **U.S. Provisional Patent Application** No. 63/799,484 (2025)

WORK EXPERIENCE

Analyst <i>JP Morgan Chase & Co.</i>	Oct 2020 – Jul 2022 Mumbai, India
– Developed an interactive dashboard for performance analysis on large high-dimensional financial datasets, ensuring statistical consistency across $\sim 100K$ records and delivering a \$1.6M annual impact. – Designed and automated ETL data pipelines for large-scale portfolio datasets, integrating internal and external feeds from 8 vendors to produce a consolidated dataset for downstream analysis. – Delivered a real-time data pipeline by integrating multiple relational databases, streaming data pipelines, and APIs, applying statistical methods to detect anomalies across 1M+ time-series records. – Technical Skills: Python, Pandas, SQL	

AWARDS AND HONORS

- **Undergraduate Research Award** in recognition of quality and extent of research in the Master's thesis (2020)
- Scholarship from the **Dutch Research Council (NWO)** for pursuing research at TU Delft, The Netherlands (2018)

PROJECTS

Visual Question Answering for Time-Series Data	Mar 2025 – Present
– Developed a Vision-Language Model (VLM)-based method for question answering on time-series data by aligning discrete token representations of signals with natural language using cross-modal alignment.	
Enhancing Conditional Diffusion of Molecules using Knowledge Graphs	Dec 2024 – Present
– Designed a novel guidance mechanism for Diffusion Models to generate targeted drugs using conditioning information from biomedical knowledge graphs, improving protein-drug docking score by $\sim 10\%$.	
Uncertainty-based Diffusion Planning for Collision Avoidance	May 2024 – Sep 2024
– Developed a novel diffusion model-based method for adaptive re-planning using uncertainty estimates from a deep ensemble to reduce collisions in a dynamically changing environment, increasing mean trajectory length by 12.7%.	

SKILLS

Languages & Libraries	Python	C/C++	SQL	Bash Shell	PyTorch	TensorFlow	JAX
MLOps & Tools	AWS SageMaker	MLFlow	Docker	Git	Hydra	LangChain	SLURM